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Controlling mechanisms for low NO_x emissions in a non-premixed stagnation point reverse flow combustor

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Abstract

The controlling processes in the operation of a non-premixed, stagnation point reverse flow (SPRF) combustor are investigated. The combustor consists of a central injector at the single open end of a cylindrical chamber. The injector inlet area is much less than the area through which exhaust products leave. The SPRF combustor operates stably at low equivalence ratios without external preheating or swirl, and produces low NO_x emissions in both premixed and non-premixed modes of operation. Non-intrusive imaging diagnostics are used to understand the combustor operation. Simultaneous Planar Laser-Induced Fluorescence (PLIF) imaging of OH radicals and chemiluminescence imaging are used to characterize the reaction and heat release zone and the flame products. Separate measurements with Particle Image Velocimetry (PIV) provide the reacting velocity field, and elastic laser sheet scattering from liquid droplets seeded into the fuel characterize its mixing. The velocity and chemiluminescence data indicate the flame is stabilized in a region of low mean and high rms velocity. Together with some product entrainment, this enables stable operation of the combustor at very lean overall equivalence ratios. The non-premixed mode of operation is found to be similar to the premixed case in many ways. Though, in non-premixed operation, the flame is lifted well away from the injector, so that significant air and fuel premixing occurs before combustion. Similar NO_x emission for both operating modes is attributed to efficient mixing of nearly all the fuel and air before burning. This is confirmed through a combination of oil droplet results, OH PLIF comparisons and laminar flame modeling. The latter indicates entrainment of products is not directly responsible for low NO_x emissions at a given overall fuel–air ratio, but rather is likely to contribute to a slight increase in NO_x levels compared to a system with no product entrainment.
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Keywords: NO_x; Non-premixed combustion; SPRF combustor

1. Introduction

Environmental concerns and legislative regulations are driving combustor manufacturers to meet increasingly more stringent emission standards while maintaining (or improving) efficiency and reliability. For example, reductions in NO_x emissions from land-based, gas turbine combustors have been achieved mainly through lean premixed/partially premixed combustion, dilution with exhaust gases and staged combustion, all of which lower peak flame temperatures and consequently NO_x emissions. In the most common approach, premixed (or partially premixed) combustion however, static and dynamic stability is compromised as the mixture is made leaner. This results from the weaker combustion process,

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which is more vulnerable to small perturbations in combustor operating conditions [1].

Flames with highly preheated combustion air can be much more stable and homogeneous as compared to room-temperature air combustion [2]. Hence the combustor can operate at much leaner equivalence ratios. Combined with the more uniform thermal field, this results in lower peak temperatures and reduced NO_x. An alternative approach to lowering NO_x is flue gas recirculation (FGR). Dilution of the reactants with *cooled*, inert products lowers the oxygen concentration and temperature before combustion, which increases ignition delay [3]. This causes the combustion zone to be spread over the entire mixing region rather than in a thin front. The increased ignition delay also allows sufficient time for premixing of fuel and air before burning occurs. The result is a more uniform temperature distribution that helps to lower NO_x. In contrast to FGR, many combustors also have some internal or external recirculation of hot products, denoted here as exhaust gas recirculation (EGR). NO_x reductions similar to those obtained by FGR have been predicted for hot EGR [4]. However, the role of EGR in lowering NO_x emissions is not completely understood, as the enhanced reactivity due to addition of hot products and radicals offsets the increased ignition delay produced by lowering the oxygen concentration.

Measurements of NO_x emissions in jet diffusion flames with coaxial air [5–7] have shown significant reduction of NO_x in hydrogen- and methane–air flames. The high shear produced by the high velocity coaxial air results in enhanced fuel–air mixing before combustion and lowers the residence times, resulting in lower emissions.

A lab-scale, ultra-low emissions combustor incorporating some of the above features was recently demonstrated [8]. This Stagnation Point Reverse Flow (SPRF) combustor's unique design enables it to burn gaseous or liquid fuels with low NO_x and CO emissions. The combustor consists of a tube with one open end and the other closed. Contrary to most combustors, the reactants and products enter and leave this combustor at the same (open) end. In the investigated configuration, the reactants are injected along the combustor center line, while the products flow in the reverse direction to exit the combustor, allowing the outflowing products and inflowing reactants to come into direct contact in a thin shear layer. A somewhat similar reverse flow geometry was employed previously for turbulent combustion studies [9]. However in that partially premixed reactor, a large recirculation zone produced by a water-cooled bluff body separated the reactant and product flows.

Because the exiting products flow directly over the reactant injector in the SPRF design, they produce "internal" preheating of the reactants before injection. While low NO_x emissions from this combustor operating with lean premixed reactants might be expected, non-premixed operation can also achieve very low NO_x levels. For example, previous work [8] showed that the NO_x emissions at a fixed equivalence ratio were roughly similar for both modes of operation.

To experimentally examine the operation of the SPRF combustor with non-premixed reactants, we employ several non-intrusive flowfield imaging methods, including planar laser-induced fluorescence (PLIF) of OH radicals, which marks the reaction zone and hot products, and chemiluminescence imaging, which indicates regions of heat release. These techniques have been commonly employed, for example to study turbulent jet flames [10,11]. Particle image velocimetry (PIV) [12] is used to characterize the velocity field of the reacting flow. Finally, scattering from oil droplets seeded into the fuel provides information on fuel mixing with air and products.

2. Experimental setup

The SPRF combustor used for the current studies is a laboratory scale, atmospheric pressure device. It consists of a dual concentric tube injector (diameter, $D_{inj} = 12.5$ mm) centrally located in a 70 mm inner diameter quartz tube, which is closed at the bottom end (Fig. 1). Premixed reactants (natural gas and air) are introduced through the injector's annulus in premixed operation. For non-premixed combustion, fuel is injected through the inner tube and air through the annulus; so no mixing can occur until the flows enter the combustor. Fuel and air flow rates are controlled separately with calibrated rotameters. The combustor is insulated with a close fitting, hollow alumina cylinder cut into four sections. For optical access, a 150° window is removed from one of the sections, which can be placed at different axial locations. Thus the optical measurements are acquired for one-quarter of the combustor at a time; the complete field is derived by splicing them together. The sheets for the laser diagnostic techniques enter the combustor through a quartz base plate. The temperature of the reactants and the exhaust gases are monitored with an unshielded K-type thermocouple. Emission measurements are obtained with a gas analyzer (Horiba PG-250) and an uncooled, ceramic sampling probe located in the exhaust gas ~25 mm into the combustor. The thermocouple and sampling probe are not present, however, while the optical measurements are recorded.

Measurements of the two-dimensional velocity field are obtained with a typical PIV system. The combustion air and fuel are separately seeded with 1–2 μm aluminum oxide particles and illuminated

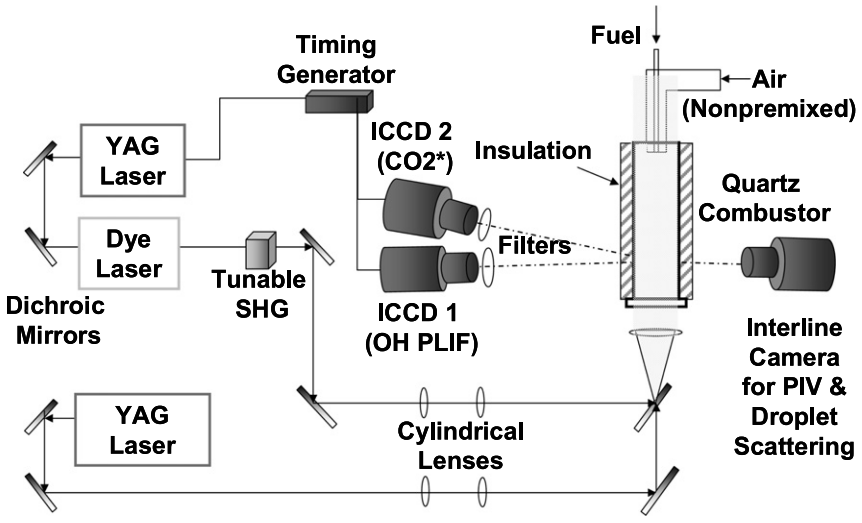

Fig. 1. Schematic of experimental set-up—OH-PLIF/Chemiluminescence and PIV data acquired non-simultaneously.

with a laser light sheet produced from the 2nd harmonic output of a dual-head, pulsed Nd:YAG laser (Surelite I-10). The beam is converted into a thin sheet (~0.4 mm) with two cylindrical lenses, and enters the combustor from the closed end (Fig. 1). For the current measurements, the laser pulse energies were nearly 90 mJ. The correlated pairs of particle scattering images are acquired with a 12-bit interline CCD camera (MicroMAX, 1300×1030 pixels) and a 50 mm, f 1/1.8 camera lens. The timing between the two laser heads and the camera shutter is synchronized with a digital pulse generator. Pulse delays between the image pairs vary from 3 to 11 μs for different regions of the combustor. The images are acquired for 55×89 mm segments of the combustor and then spliced together to produce the complete combustor velocity field. Commercial software (Insight 6, TSI) is used to calculate the velocity field. An FFT based cross-correlation technique is employed to find the average particle displacement in a 64×64 pixel region with a 50% overlap in the interrogation areas yielding a total of 1209 vectors. The particle displacements are obtained with a 0.1 pixel accuracy using a Gaussian sub-pixel interpolation algorithm. Each vector represents the average velocity in an interrogation volume $2.19 \times 2.19 \times 0.4$ mm.

Mixing of unburned, cold fuel with air and hot products is investigated by seeding the fuel with 5–10 μm diameter olive oil droplets (and without alumina seeding of either fuel or air). The oil droplets are generated with a laskin nozzle seeder design [13], where the nozzle is immersed in the oil. The number of droplets produced from this device remains linear with input pressure, as long as the nozzles are choked. The distribution of droplets in the combustor is determined with the same optical system employed for PIV.

OH fields are determined with a PLIF system tuned to excite the $Q_1(6)$ line of the $A^2\Sigma^+(v' = 1) \leftarrow X^2\Pi(v'' = 0)$. The excitation radiation is produced by frequency-doubling the output of a tunable (Lambda Physik) dye laser, pumped by the second harmonic of a Nd:YAG. The laser beam, with a pulse energy of roughly 17 mJ, is expanded into a sheet 69 mm wide and 300 μm thick with three fused silica lenses. The laser sheet passes through a diametrical plane, and illuminates nearly the whole width and length of the combustor. The emitted fluorescence light is detected normal to the sheet with a 25 mm intensified camera (PI-MAX, 1024×256 pixels) equipped with a UV-Nikkor lens (105 mm, f/4.5). Schott glass filters (WG305, UG11) placed in front of the lens limit the detected fluorescence to the range 300–370 nm. A pixel resolution of ~300 μm is obtained with this configuration.

A second intensified camera (Princeton Instruments ICCD-576-S/RB-E, 18 mm intensifier, 384×576 pixels) and interference filter (430 ± 5 nm) is used to collect CH^* and CO_2^* chemiluminescence. The intensifier gate of the second camera is synchronized with the OH-PLIF system, such that the exposure begins 100 ns after the OH laser (after the OH fluorescence has decayed) and lasts 250 μs. This provides nearly simultaneous measurement of OH and chemiluminescence. The chemiluminescence camera is also mounted at right angles to the laser sheet, and its field-of-view covers the full width of the combustor, resulting in a pixel resolution of ~250 μm.

3. Results and discussion

As noted previously, one of the interesting features of the SPRF configuration is its ability to

achieve essentially premixed combustor performance in a non-premixed mode of operation. Figure 2 shows a comparison of NO_x emissions for the current configuration operating in premixed and non-premixed modes. The non-premixed emissions are comparable to those obtained in premixed operation across a wide range of conditions. Possible explanations are that the fuel and air are completely premixed within the combustor, before burning, or that the entrainment of products into the fuel and air (essentially EGR) acts to lower O-atom concentrations thereby reducing NO_x levels. To better understand the non-premixed operation of the combustor, velocity fields and flame structure were determined for a total (fuel + air) inlet mass flow rate of 8.1 g/s and an overall equivalence ratio (ϕ) of 0.58 corresponding to a power density of 13.3 MW/m^3. These conditions were chosen to match a separate study of the combustor operating in premixed mode.

The average axial velocity field (Fig. 3a) based on 400 individual realizations, indicates peak air velocities of the order of 90 m/s. This is higher than the calculated value based on the cold, reactant inlet flow. However, preheating of the incoming air by the exhaust products flowing over the outside of the injector will tend to increase the inlet velocity. Since the inner fuel tube is shielded by the air, the fuel has little preheating. The closed end opposite the injector forces the incoming jet to slow down and reverse direction. The reverse flow velocities of the product gases are around 20 m/s. On average, there is also noticeable recirculation between the forward (jet) and reverse flows for $x > 75$ mm, where x denotes axial distance from the injector exit.

The mean velocities and turbulence intensities are compared for premixed and non-premixed modes in Fig. 3b. Near the open end of the combustor ($x < 100$ mm), the centerline axial velocities are higher for premixed combustion.

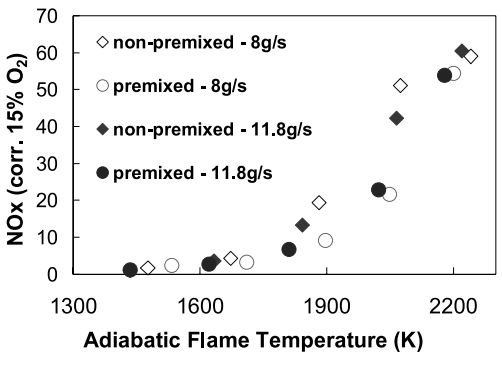

Fig. 2. Variation of NO_x emissions with adiabatic flame temperature based on overall equivalence ratio and measured inlet temperature in premixed and non-premixed modes.

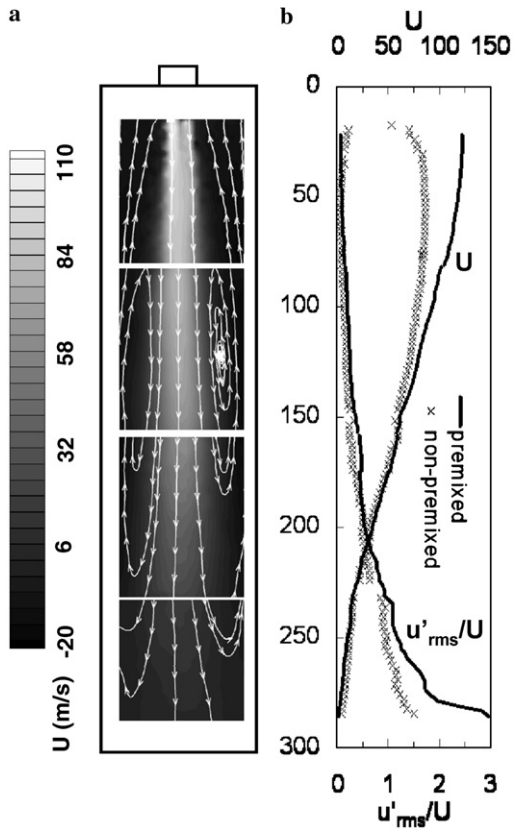

Fig. 3. (a) Mean axial velocity contours and streamlines in non-premixed mode. (b) Centerline variation of mean axial velocity and turbulence intensity for premixed and non-premixed modes of operation.

Initially, some difference would be expected due to the difference in inlet area; premixed operation utilizes only the annular passage of the concentric injector. However, it is unlikely that this difference would persist more than a few injector diameters. Further downstream, the axial velocities are approximately equal, with the premixed mode exhibiting greater turbulence levels. The overall flow features, however, are similar in both premixed and non-premixed modes of operation with the presence of a low U and high u'_{rms} region in the second half of the combustor that has been shown to help stabilize the flame for premixed operation in a separate study.

The importance of this region for non-premixed operation is evident from the average (100 images) chemiluminescence field (Fig. 4a). Most of the heat release occurs for $150 < x < 220$ mm. Unlike the premixed combustor (Fig. 4b), there is negligible heat release close to the injector; the flame is completely lifted and stabilized only in the region of lower velocity and high unsteadiness. The large standoff distance suggests there is time

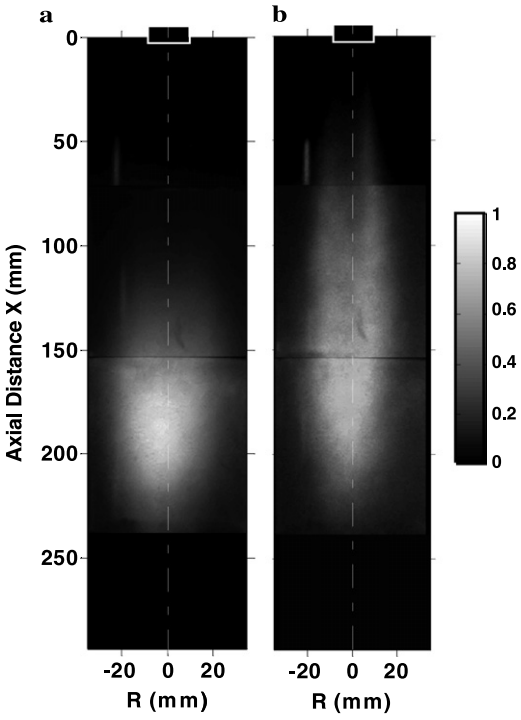

Fig. 4. Comparison of average chemiluminescence fields for (a) non-premixed and (b) premixed modes of operation.

for fuel and air to mix, and also to entrain product gases before reaching the flame zone. The reduced heat release in the first half of the combustor also explains the lower initial velocities found for non-premixed operation (Fig. 3); the lower heat release causes less expansion and a reduced acceleration of the flow.

The rate of fuel mixing was studied by imaging light scattered from oil droplets seeded into the fuel as described in the experimental section. The boiling point of olive oil is 570 K; hence the droplet size and therefore the scattered intensity is reduced either by: (1) evaporation caused by partial mixing with hot products or proximity to the flame zone, or (2) dilution due to mixing with air or products. Mixing with air that has not mixed with products will not lead to significant droplet evaporation, since the inlet air temperature is below 450 K. For an average droplet size of 7 μm and for fuel velocities above 50 m/s, it is estimated that a rapid drop in scattering intensity due to evaporation would occur for temperatures greater than ~900 K. Hence, it may be inferred that the regions where droplets are observed represent locations where the local temperature does not exceed ~900–1100 K.

Figure 5 shows the variation of the normalized average, centerline droplet intensity with axial distance from the injector exit. Also shown is the average chemiluminescence intensity along the centerline. The decrease in droplet scattering intensity is less than 20% within the first 40 mm ($\sim 3D_{\mathrm{inj}}$). The corresponding chemiluminescence intensities in this region are essentially zero. These observations suggest some fuel–air mixing occurs here, but little entrainment of products into the fuel stream. In other words, the fuel is initially shielded from the hot products by the annular air.

Further downstream, the scattered intensities drop nearly linearly, until $x \sim 125$ mm, at which point the droplet scattering is reduced to the local background level. The nearly linear decrease in droplet scattering before this point suggests significant dilution of the fuel with air, and possibly products, is occurring, but that the fraction of products is sufficiently low that the temperatures do not exceed the 900–1100 K level much before $x \sim 125$ mm. The rapid increase in average chemiluminescence (or heat release) for the region beyond 125 mm indicates temperatures there are high enough to evaporate the oil droplets.

These results suggest that fuel–air mixing occurs in the initial zone of the combustor, while air and product mixing occurs in the shear layer between the forward going reactants and the reverse flow of products. Though not shown here, the instantaneous velocity fields indicate the presence of large vortical structures in the region of the combustor where the chemiluminescence rapidly rises. These vortices cause rapid mixing of partially premixed fuel–air packets with product or product-air packets. Therefore, it may be inferred that the fuel and air are at least partially premixed before combustion occurs, and there is some amount of products mixed with the fuel and air when it burns. This product entrainment facilitates the stabilization of lean flames in the downstream region characterized by low average velocities and high turbulence levels. At this stage, it is still unclear whether the comparable NO_x

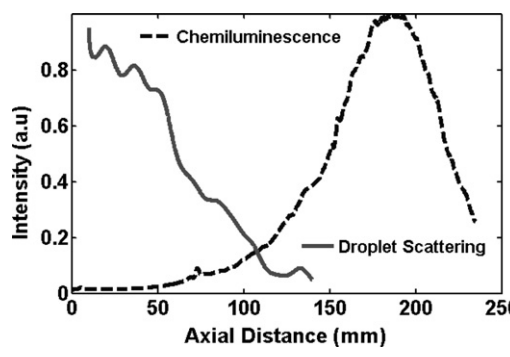

Fig. 5. Variation of centerline droplet intensity and average centerline chemiluminescence signal with axial distance from injector.

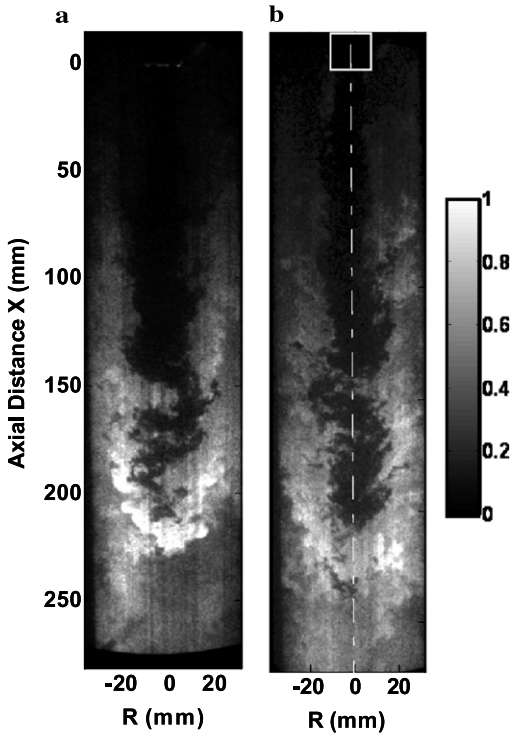

Fig. 6. Instantaneous OH PLIF field for flow rate = 8.1 g/s, $\Phi = 0.58$ (a) non-premixed (b) premixed.

levels between premixed and non-premixed modes is a result of product dilution or fuel–air premixing to the overall equivalence ratio of 0.58 before burning occurs.

This question is addressed by comparing the OH fields for the two modes of operation. Figure 6 shows the instantaneous OH fields for premixed and non-premixed operation. Starting at $x \sim 110$ mm, the reactant streams (the central dark regions in the images) show similar structures and penetration for both premixed and non-premixed cases. However, just beyond the point of maximum penetration, where the PLIF images show somewhat uniform signals over a broad region, the non-premixed case produces higher fluorescence intensities. Comparing 100 instantaneous images for each case, the non-premixed intensities are approximately 40% higher. Based on the assumption that these broad uniform regions are close to OH equilibrium, the variation of equilibrium OH concentration and the corresponding (simulated) PLIF signal, which also takes into account variations in quenching and population fraction, were calculated as a function of equivalence ratio. From the simulation results (Fig. 7), it is seen that a 40% increase in PLIF signal would indicate that the flame in the non-premixed mode is burning only slightly richer ($\phi \sim 0.615$) than in the premixed case ($\phi = 0.58$).

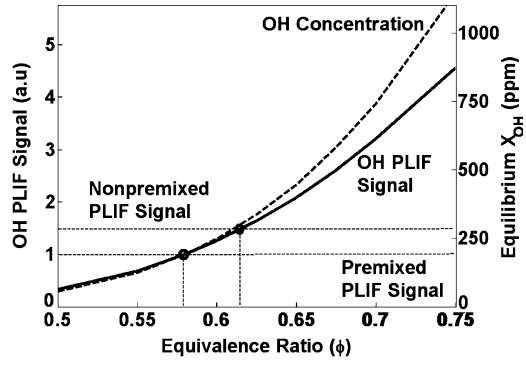

Fig. 7. Variation of equilibrium OH concentration and PLIF signal with equivalence ratio.

This marginal increase in equivalence ratio of the flame above the overall value indicates effective mixing of nearly all the fuel and air before they reach the flame zone. This would also explain the similarities in NO_x emissions for the two combustor modes. The above analysis is valid even if there are products entrained with the reactants, as the equilibrium OH concentrations would not be affected by recirculation of adiabatic products.

To explore the alternative, that recirculation of hot product gases has an effect on NO_x emissions at a given equivalence ratio, laminar one-dimensional flame simulations (CHEMKIN, GRIMech 3.0) were carried out. The effect of hot product gas recirculation was simulated by adding a fraction of hot products ($f_P \equiv$ product mass fraction in the reactants) back into the inlet reactant stream. The temperature and composition of the adiabatic products was based on results for $f_P = 0$. Results were obtained at two initial reactant temperatures (before mixing) of 300 and 600 K. The measured entering air temperature is between these values. The simulated emission levels (Fig. 8) correspond to the NO_x concentration just after the flame front, i.e., where the heat release has dropped to 1% of its peak value. The adiabatic flame temperatures for the two cases are 1630 and 1860 K respectively. At these relatively low flame temperatures, the amount of thermal NO_x produced downstream is expected to be low.

In both cases, EGR of adiabatic products appears to only slightly increase NO_x emissions. For an initial reactant temperature of 300 K, corrected NO_x emissions (at 15% O_2) obtained for different product fractions show little variation with increasing f_P. For the elevated preheat temperature of 600 K and with 50% of the preflame mixture being products ($f_P = 1$), the NO_x emission increase from $\sim$2 to 3 ppm. Hence, moderate amounts of product entrainment in the non-premixed SPRF combustor would not reduce NO_x compared to premixed operation, and should only increase it by a small amount. To compare the

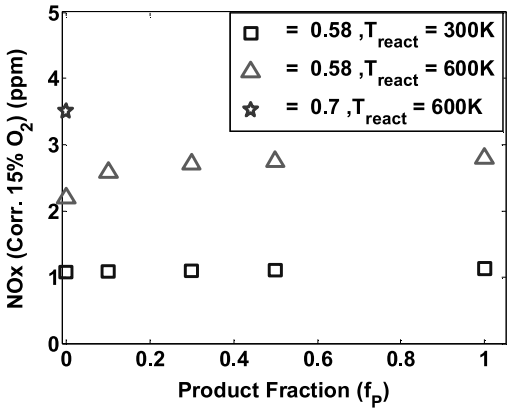

Fig. 8. Effect of adiabatic exhaust gas recirculation on flame NO_x from laminar flame calculations.

effect of only partial premixing of fuel and air, Fig. 8 includes a third case, with reactant equivalence ratio of 0.7 and a preheat temperature of 600 K, but no product entrainment. The flame NO_x increases to 3.5 ppm, and since the adiabatic flame temperature of this condition is over 2000 K, some added thermal NO_x would be likely. This supports the earlier conclusion that nearly all the fuel and air are mixed before they reach the flame zone when the SPRF combustor operates in non-premixed mode. Otherwise, we would expect higher (corrected) NO_x emissions.

4. Conclusions

The flowfield and flame structure of a stagnation point, reverse flow combustor operating were investigated for overall lean fuel–air mixtures at atmospheric pressure. Emission measurements performed on this combustor indicate that comparable, low NO_x emission levels are achieved in both non-premixed and premixed modes of operation. Various optical diagnostic techniques were employed to explain these similarities.

The lower reactant velocities for non-premixed operation in the region of the combustor close to the injector, and the lack of chemiluminescence emissions from this area indicate little heat release occurs here. This is in contrast to premixed operation, which shows a flame stabilized in the near field shear layer. Measurements of the non-premixed velocity field confirm the existence of a region with low average velocity but high turbulence in the lower half of the combustor. This region is also the source of most of the chemiluminescence emissions (heat release), and thus represents the flame stabilization zone for non-premixed operation. Instantaneous velocity and (non-simultaneous) OH PLIF data point to the presence of large vortical structures in the midsection of the combustor that can entrain hot combustion products into the reactant stream, and thereby increase the chemical rates and aid stabilization. Thus stabilization is a combined result of product entrainment, low U and high u'_{rms}. Since the low velocity region with high product entrainment must exist for most inlet flow rates, the SPRF geometry can produce stable operation even at high loadings and for very lean mixtures.

Similarity in NO_x emissions for both operating modes is attributed to efficient mixing of nearly all the fuel and air before burning. Since the flame is lifted well away from the injector, air and fuel can premix before combustion occurs. This is confirmed by the oil droplet scattering results, which also show that hot products are initially shielded from the fuel by the surrounding air flow. Comparisons of OH PLIF results for non-premixed and premixed operation with equilibrium calculations indicate the flame in non-premixed operation is burning only slightly richer than in the premixed case. It is also shown with the help of laminar flame calculations that the entrainment of products is not directly responsible for the low NO_x emissions at a given overall fuel–air ratio, but rather, is likely to contribute to a slight increase in NO_x levels compared to a system with no product entrainment.

In summary, the SPRF geometry produces a stable flame even at lean operating conditions, with NO_x emissions for non-premixed operation that are nearly as low as those produced with premixed reactants. The low emission levels are attributed to the shielding of the fuel from the hot products, and efficient mixing of most of the fuel and air (and some products) before combustion occurs. For applications at elevated pressures, it is anticipated that the flow structure of the SPRF combustor will remain similar, though increased pressure will increase chemical kinetic rates and influence the balance between chemical and flow times.

Acknowledgments

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Comments

Shigeru Hayashi, Japan Aerospace Exploration Agency, Japan. Did you compare the measured CO concentrations with the equilibrium ones to estimate the magnitude of lost heat?

Please comment on the distribution of local "dilution ratio," i.e., the fraction of burned gas to total mixture (burned gas + fuel + air) for the premixed case if you measured the fresh mixture.

Reply. The measured CO concentrations are lower than equilibrium values and this is mainly due to recombination of CO in the gas analyzer probe. Separate measurements of product gas temperatures at the exit of the combustor have been carried out and results indicate that the magnitude of heat losses in the SPRF combustor is less than 10%.

The "dilution ratio" or product fraction has not been measured to date; however, work is currently underway to measure this quantity.

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Bassam Dally, The University of Adelaide, Australia. Your flames look similar to those achieved under MILD combustion. You argue that the exhaust gases do not play any role in the low NO_x levels, would this imply that you will be able to have an open ended channel and achieve the same results? Can you comment what is the effect of exhaust gases then?

Reply. The presence of a closed end facilitates entrainment of the returning product gases. The main role of internal product gas recirculation in the SPRF combustor is to aid in stabilization of the flame at very lean equivalence ratios thereby resulting in low NO_x emissions.